

Luis San Martin

Computer Vision Engineer · Software Developer

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PROFILE

I train and deploy **computer vision models** on cameras in real operation, and build the data systems and **agents** that report unattended. Over three years with software in production, working directly with the owner of each operation.

EXPERIENCE

Software Developer — *Distribuidora El Maravilloso*

Nov 2025 – Present · Concepcion, Chile

- Integrated the point-of-sale system — Firebird, **no API** — into cloud PostgreSQL: today **11,900 sales and 40,000 product lines**, ~78 per day.
- Unattended ETL** that pulls tax documents from the national tax authority portal and computes the monthly tax indicators: **305 purchase invoices** imported and classified across 87 suppliers.
- Data quality control**: cross-checking three sources — tax documents, electronic payments and point-of-sale records — to detect duplicated income and till mismatches: **283 alerts** raised without altering the client's figures.
- Agents in operation** over WhatsApp and Telegram: sales close, tax indicators and threshold alerts, with a **custom intent classifier** that avoids depending on an external API.
- Real-time reporting and monthly executive report. Diagnosed and fixed **data integrity incidents in production** — record loss, clock drift — recovering the data.

Stack: Python · SQL · PostgreSQL · Firebird · JavaScript · Linux · scheduled jobs · Langfuse / OpenTelemetry on the AI routes

Software Developer — *LetsFit Company* · independent engagement

Feb 2023 – Present · Remote

- Migrated the business off their commercial platform — **~170 active members** —: extracted the full database by **web scraping** and moved clients and history with no loss.
- Designed the **PostgreSQL** schema: multi-site with row-level security and versioned migrations, for **2 locations**.
- Built the admin panel and its reporting: attendance, occupancy by site and time slot, revenue, and commission settlement for **14 professionals** across a 768-session schedule.

Stack: PostgreSQL · SQL · Python · TypeScript

Computer Vision Engineer — *St. Andrews · Mussels Chiloé*

Jun 2024 – Jul 2025 · Chiloé, Chile

- Multi-camera **detection, classification and counting** system across 3 packing stations on a line handling **13,000 kg per day**, at the world's largest mussel producer.
- Multi-object tracking (SORT)** to hold each package's identifier across frames and prevent double counting when the belt stops or an operator reaches in.
- Larvae counting model** that cut manual analysis from **45 minutes to seconds**: 696 individuals in 2.93 s, trained on a **custom dataset annotated inside the plant**.
- Robust to changing lighting and wet surfaces. **Ran a full season with no manual intervention**, with inference on the plant's own machines.

Stack: Python · PyTorch · YOLOv8 · OpenCV · SORT · RTSP over IP cameras

COMPUTER VISION & AI PROJECTS

Helmet detection on an embedded camera — *Exwood · electric mobility*

2026

- Fine-tuned YOLOv8** to 3 classes (scooter, helmet, no-helmet) over **12,321 images** and 80 epochs: **mAP@50 of 85.5%**, precision 85.0% and recall 80.9%.
- Capture on an **embedded XIAO ESP32-S3 Sense camera** —not a CCTV camera— with **inference on a local GPU** at 9.6 frames per second: the constraint was that the hardware had to be mountable on the scooter itself.

Stack: PyTorch · YOLOv8 · OpenCV · Python · ESP32-S3

Retail footfall analytics — *with client, in production*

Feb 2026 – Present

- Entry and exit counting on the CCTV cameras the store already had, cumulative traffic heatmaps and configurable line-crossing alerts, with no additional hardware.
- Also a GPU-accelerated variant of the engine on **DeepStream** (pyservicemaker): NVDEC decode, YOLO batched on **TensorRT** and nvtracker, to scale across several cameras in parallel.

Stack: Python · YOLOv8 · OpenCV · DeepStream · TensorRT

Personal protective equipment (PPE) detection — *final-year project* 2026 — ongoing

— Detection of helmets, vests and other PPE on industrial sites. The core contribution is the **custom dataset annotated on a real work site**, rather than public banks that do not represent actual working conditions.

Stack: PyTorch · YOLOv8

TECHNICAL SKILLS

Computer vision	PyTorch · YOLOv8 · OpenCV · object detection and classification · multi-object tracking (SORT) · fine-tuning · custom datasets annotated in the field · real-time inference
Training & deployment	Cloud training (Colab) · on-premise inference on the client's GPU (DeepStream · TensorRT · NVDEC) · edge (ESP32-S3) · RTSP · evaluation by mAP, precision and recall
Agents & generative AI	Conversational agents on WhatsApp and Telegram · LLM APIs (Groq, Anthropic) · intent classification · custom n-gram classifier with no API · LLM observability with Langfuse and OpenTelemetry
Languages	Python · SQL · TypeScript / JavaScript · C++ (ESP32 firmware)
Data & reporting	PostgreSQL (modelling, versioned migrations, row-level security) · Firebird · SQLite · ETL with no API · web scraping · dashboards · threshold alerts
Infrastructure	Linux · Docker · Git · GitHub Actions · cloud servers · scheduled jobs and monitoring

EDUCATION

Automation Engineering	— <i>Universidad del Bío-Bío</i>	Graduated 2024 · Degree completion 2026
Languages	Spanish (native) · English (technical reading)	
Availability	Immediate · Concepcion, Chile — remote or hybrid	

Case studies with images and metrics at [sanmaglass.github.io](#). Repositories are private as they hold client production code; available for review under agreement.